Doubt Session

27 March 2026 20:11

Q. 2→Binary addition are → decimal.

→ 1) Write a function to return a decimal number, given binary number as an input.

Q - Sum of prime factors.

```
Public Class Main {
    public Static void main() {
        - - - -
    }
```

```
public Static int bin2dec(int bin){
    int num = 0;
    - - - -
    - - - -
    - - - -
}
//Sout(num);
return num;
}
```

//without using power function.

```
public int BinarytoDecimal(int i){
    int dec = 0;
    int rem;
    int power =1;
    while(i>0){
        rem = i%10;
        dec += (rem * power);
        i = i/10;
        power *= 2;
    }
    return dec;
}
>
```

→ Integer.parseInt();

```
public void main() {
    int a = 1000100; b = 0111;
    int dec1 = BinarytoDecimal(a);
                    Class
    int dec2 = ~~~(b);
    Sout(dec1 + dec2);
```

Q. write a function to return factorial of a number;

```
int fact = 1;
for(int i = 1; i <= n; i++) {
fact = fact * i;
}
return fact;
}
```

n=0;

i=1 ; i<=0

→ Pascal triangle

$\boxed{{}^{row}C_{col}}$

$\boxed{{}^{row-1}C_{col-1}}$

row=0 ↓ col 0
1
↓ col 1
row=1 1 1 ↓ col 2
row=2 1 2 1
row=3 1 3 3 1
1 4 6 4 1

$^{2}C_{1} = 2$

$^{3}C_{2} = 3$

$$\boxed{^{n}C_{r} = \frac{n!}{(n-r)! \times r!}}$$

0
1
2
row = 3

row = 3 0 1 2 3
O/p → 1 3 3 1

n ~~1 2~~ 3

```
int n = 3;
for (int j = 0; j <= n; j++){
    Sout((fact(n))/(fact(i) * (fact(n-i)));
}
```

1
1 1
1 2 1
1 3 3 1

i = 0

fact(0) → 0

fact(0) → 1

1
1 1
1 2 1

row = 3
row = 0 →

r

row = 0 → n (1, 2, 3 ......)

n → i

```
for(int i=0; i<=n; i++){
    for(int j=0; j<=i; j++){
        Sout((fact(i)/(fact(j)*fact(i-j)));
    }
    Sout("\n");
}
```

j = 0

i=0   1

i=1

Q. Decimal to binary Conversion.

Decimal 29 → binary

2)29(14
28
1

rem

2	29	1
2	14	0
2	7	1
2	3	1
	1	

1 1 1 0 1

binary 11001 → decimal

Bin 46 →

2	46	0
2	23	1
2	11	1
2	5	1
2	2	0
	1	

2)46(23
46
0

2)23(11
22
1

1 0 1 1 1 0

```
int ans = 0;
while(num>0){
    int rem = num%2;
    ans = ans*10 + rem;
    num = num/2;
}
```

Seems fine!

Mistake

Dry run
8

limits code, // rem = 0   ans=0000

num =

}

//reverse the digits code,

```
int rev=0;
while(ans>0){
    int rem = ans%10;
    rev=rev*10+rem;
    ans =ans/10;
}
sout(rev);
```

00

//

rem = 0
ans = 0

num = 4
rem = 0
ans = 00

String

ans = 0

ans=0000
= 0

num = 2
rem = 0
ans=000

num=1
rem = 1
ans = 1

1

1000

0001

n=8

2	8	0
2	4	0
2	2	0
	1	

$\boxed{1000}$

1000

String

"0001"

rev the String